

Test Plan Document

1. Project Overview

This project focuses on designing and implementing an automated testing framework for web applications to ensure end-to-end validation through manual, automation, API, database, and performance testing.

2. Project Description

The project includes:

- Functional and exploratory testing
- Automation using Page Object Model (POM)
- API validation using Postman
- Database verification using SQL
- Performance testing using JMeter
- CI/CD integration using Jenkins

3. Tested Applications

SauceDemo (Primary Website)

URL: <https://www.saucedemo.com>

Type: E-commerce

Purpose: Manual testing and selenium Automation Testing

ReqRes

URL: <https://reqres.in>

Purpose: API Testing (Postman/Rest Assured)

OpenCart Demo

URL: <https://demo.opencart.com>

Type: E-commerce

Purpose: Database Testing and data validation using SQL

4. Tools Used

- Selenium
- Postman & Newman
- MySQL
- JMeter
- Jenkins
- Jira
- Git/GitHub
- Cucumber

5. Test Deliverables

- Test Plan Document
- Test Cases (Manual & Automated)
- Bug Reports
- Automation Scripts
- API Collections
- Performance Reports
- Final Test Report

6. Entry Criteria

- Test environment is ready
- Applications are accessible
- Test data is prepared
- Tools are configured

7. Exit Criteria

- All test cases are executed
- Critical defects are fixed
- Test results are documented
- Final report is completed

8. Test Schedule (Timelines per Sprint)

Sprint 1: Planning

- Define project scope and objectives
- Prepare Test Plan document
- Define Entry and Exit Criteria
- Plan testing schedule and resources

Deliverables: Test Plan, Project Roadmap

Sprint 2: Analysis & Design

- Perform exploratory testing to understand application behavior
- Design test cases for UI, API, and database
- Identify automation candidates
- Create BDD feature files

Deliverables: Test Cases, Feature Files, Initial Framework Design

Sprint 3: Implementation

- Develop automation scripts using Selenium
- Execute API testing using Postman
- Write SQL queries for database validation
- Setup GitHub repository and version control

Deliverables: Automation Scripts, API Collections, SQL Queries

Sprint 4: Execution & Completion

- Execute all manual and automated test cases
- Perform performance testing using JMeter
- Analyze results and generate reports
- Integrate CI/CD pipeline using Jenkins

Deliverables: Test Reports, Performance Reports, Final Documentation

9. Agile Practices

Daily Standups: Short daily meetings to track progress and blockers.

Sprint Retrospective.